

Network Operations Strategy Report

Graph-Based ETA Intelligence for Delhivery Logistics Network

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Executive Summary

Accurate Estimated Time of Arrival (ETA) prediction is a critical component of modern logistics operations. In large transportation networks, delays often arise not only from route distance and travel time but also from structural bottlenecks within the logistics network. Traditional ETA estimation methods fail to account for these network-level effects, resulting in inaccurate delivery commitments and increased SLA breaches.

This project develops a Graph-Enhanced ETA Prediction Framework for Delivery's logistics network. The transportation system was modeled as a directed graph where logistics hubs represent nodes and shipment movements represent edges. Graph analytics were combined with machine learning to identify critical bottlenecks, quantify corridor-level risks, and improve ETA prediction accuracy.

The proposed framework achieved a significant improvement in predictive performance, reducing Mean Absolute Error (MAE) from 47.12 minutes to 31.49 minutes, representing a 33.17% improvement over the baseline model.

Dataset Overview

The analysis was conducted using shipment movement records across Delhivery's logistics network.

Dataset Statistics

Metric	Value
Raw Shipment Records	144,867
Unique Trips	14,817
OD Legs	26,368
Source Hubs	1,508
Destination Hubs	1,481

The dataset contains information related to:

- Shipment movements
- Origin and destination hubs
- Actual delivery times
- OSRM estimated travel times
- Travel distances
- Route categories
- Timestamp information

These variables were used to construct both operational and network-based intelligence features.

Logistics Network Construction

A directed transportation network was created using shipment flow data.

Network Statistics

Metric	Value
Nodes (Hubs)	1,590
Edges (Corridors)	2,508

Each node represents a logistics hub while each edge represents a transportation corridor connecting two hubs.

Graph analytics were performed using NetworkX to compute:

- Degree Centrality
- Betweenness Centrality
- In-Degree
- Out-Degree

These metrics help identify hubs that play a critical role in overall network connectivity and delay propagation.

Bottleneck Hub Analysis

Betweenness centrality was used as the primary indicator of network importance because disruptions at high-centrality hubs can impact a large number of downstream routes.

Top Bottleneck Hubs

Hub ID	Betweenness Centrality
IND000000ACB	0.234
IND562132AAA	0.099
IND712311AAA	0.092
IND501359AAE	0.067
IND160002AAC	0.057

Key Observation

A small number of hubs account for a disproportionately large share of network traffic. These hubs act as critical transit points and contribute significantly to network-wide delays when congestion occurs.

Operational improvements at these locations are expected to produce benefits across multiple delivery corridors simultaneously.

Graph-Based Feature Engineering

Traditional ETA models primarily rely on route distance and travel time. To capture network intelligence, additional graph-derived features were engineered.

Graph Features

- Source Hub Centrality
- Destination Hub Centrality
- Corridor Delay Ratio
- Corridor Traffic Volume
- Network Risk Score
- In-Degree and Out-Degree Metrics

Temporal Features

- Trip Hour
- Day of Week
- Month

Operational Features

- OSRM Time
- OSRM Distance
- Actual Distance to Destination

- Route Type

These features capture both local route characteristics and broader network behavior.

Machine Learning Framework

An XGBoost Regressor was selected due to its ability to model nonlinear relationships and handle heterogeneous feature types.

Two models were evaluated:

Baseline Model

Features:

- OSRM Time
- OSRM Distance
- Distance to Destination
- Route Type

Graph-Enhanced Model

Features:

- All baseline features
 - Network centrality measures
 - Corridor delay metrics
 - Network risk indicators
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Model Performance

Baseline Model

Metric	Value
MAE	47.12
RMSE	108.40
R ²	0.9297

Graph-Enhanced Model

Metric	Value
MAE	31.49
RMSE	90.92
R ²	0.9506
Accuracy @ 15%	54.46%

Improvement Achieved

Metric	Improvement
MAE Reduction	33.17%
RMSE Reduction	16.13%
R ² Improvement	+0.0209

The graph-enhanced framework substantially outperformed the baseline model, demonstrating the value of network-aware ETA prediction.

Feature Importance Insights

The most influential predictors were:

Feature	Importance
OSRM Distance	0.523
OSRM Time	0.244
Actual Distance to Destination	0.102
Corridor Delay Ratio	0.047
Route Type	0.013

A notable finding is that Corridor Delay Ratio emerged as one of the most important predictors, validating the contribution of graph-based intelligence to ETA estimation.

Operational Recommendations

Based on the network analysis and model findings, the following actions are recommended:

1. Prioritize Critical Hubs

Allocate additional processing capacity and monitoring resources to:

- IND000000ACB
- IND562132AAA
- IND712311AAA

These hubs exhibit the highest network influence and operational risk.

2. Monitor High-Risk Corridors

Implement automated alerting systems for corridors exhibiting elevated delay ratios.

3. Integrate Graph Risk Scores

Use network risk indicators directly within operational ETA systems to improve customer-facing predictions.

4. Infrastructure Planning

Future investments should prioritize hubs with high betweenness centrality, as improvements at these locations yield the greatest network-wide impact.

Business Impact

The proposed framework offers several operational benefits:

- Improved ETA accuracy
- Reduced SLA breaches
- Better customer experience
- Enhanced visibility into network bottlenecks
- Data-driven infrastructure planning
- Improved logistics decision-making

The achieved 33.17% reduction in prediction error demonstrates the practical value of combining graph analytics with machine learning in logistics operations.

Conclusion

This project successfully developed a Graph-Enhanced ETA Prediction System for logistics operations using network science and machine learning. A transportation graph consisting of 1,590 hubs and 2,508 corridors was analyzed to identify bottlenecks and quantify operational risk.

By integrating graph-derived features into an XGBoost framework, ETA prediction error was reduced from 47.12 minutes to 31.49 minutes while achieving an R^2 score of 0.9506. The results demonstrate that network-aware intelligence can significantly improve ETA prediction performance and provide actionable insights for logistics operations.

The framework can serve as a foundation for future real-time ETA systems, dynamic routing strategies, and network optimization initiatives.